



JAY KRISELL
Electrical Engineer

CONTACT

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LINKS

[LinkedIn Profile](#)
[GitHub](#) — [SpaceCadetJ](#)
[Portfolio Website](#)

INTERESTS

- Embedded Control Systems
- Reinforcement Learning
- Motor Control & FOC
- Motion Planning
- PCB Design & Fabrication
- Robotics Commercialization

REFERENCES

Dr. Tuan Dang
Asst. Professor, EECS
University of Arkansas

Mike Kieffer
VP, Keenfinity
612-237-6522

SFC Brandon Dostert
Readiness NCO, 1-185th C-Co
479-653-6110

ABOUT ME

I am a graduate researcher at the University of Arkansas and the founder of Kinetic Intelligence Labs. My work is focused on reinforcement learning for robotic control and developing the hardware to support it. I am building motor controllers and training systems for robotic actuators because I think the hardware side of robotics needs to catch up with what the software side is already capable of.

Before grad school I spent six years in the Army as an aviation electrician, and that experience shaped how I think about engineering. I care about building things that are reliable, testable, and actually work in the real world. I bring that same approach to everything I do now, whether it is research in the lab or developing a product for market.

WORK EXPERIENCE

Cognitive Robotics Lab — University of Arkansas — Fayetteville, AR

Graduate Research Assistant

Jan 2025 – Present

- Conducting reinforcement learning research under Dr. Tuan Dang, comparing PPO and AWR policy optimization for motion imitation in humanoid robots.
- Deploying simulation environments in NVIDIA IsaacGym and MimicKit for physics-based motion capture replay and control policy training.
- Developing custom reward functions and training pipelines for robotic actuator control using PyTorch-based RL frameworks.

U.S. Army National Guard — Camp Robinson, AR

Aviation Electrician (15F) — Sergeant

Dec 2019 – Dec 2025

- Led avionics maintenance on UH-60 and UH-72 airframes: sensors, communications, imaging systems, IVHMS, and flight instruments.
- Supervised maintenance teams and maintained readiness documentation as non-commissioned officer.
- Completed 7-month AIT at JBLE Fort Eustis covering UH-60 and CH-47 avionics. Graduated with aviation maintainer certification.

Bosch (Electro-Voice R&D) — Burnsville, MN

Quality Engineer Co-Op

Jul 2023 – Dec 2023

- Designed and validated prototype electronics and automated reliability tests for professional loudspeakers.
- Conducted DSP, endurance, stress, and frequency response testing in anechoic chambers using laser vibrometry.
- Coordinated with customers on purchase orders and quality constraints for developing and competitive products.

EDUCATION

University of Arkansas — 2025 – Present

M.S. Electrical Engineering (In Progress)

Cognitive Robotics Lab · Reinforcement learning, robotic control, embedded intelligence

University of Arkansas — 2021 – 2025

B.S. Electrical Engineering

Circuit design, robotics, control systems

JBLE Fort Eustis — U.S. Army — 2020 – 2021

Helicopter Avionics AIT (15F)

7-month advanced training · Aviation maintainer certification

University of Arkansas — Little Rock — 2018 – 2020

Associate of Arts

Concurrent enrollment · Graduated high school top 5%

CORE SKILLS

- Python · PyTorch · RL
- C/C++ · Embedded
- MATLAB · Simulink
- ROS2 · Gazebo · IsaacGym
- KiCad · Cadence Allegro
- STM32 · ARM Cortex-M
- FOC / BLDC Motor Control
- PCB Design (6-layer)
- CAN Bus · LoRa · SPI/I2C
- SMD Soldering · Rework
- Git · Linux · Docker
- Circuit Troubleshooting

COURSEWORK

- Cognitive Robotics
- Robot Localization (KF/EKF)
- Control Systems
- Electromagnetics
- Energy Systems
- Signals & Systems

CERTIFICATIONS

- FAA Aviation Maintainer
- Army 15F MOS Qualified
- Secret Security Clearance

LEADERSHIP

Six years as an Army NCO leading avionics maintenance teams under strict safety and readiness standards. Cross-functional coordination at Bosch R&D. Founded and operating an early-stage deep-tech robotics venture.

PROJECTS & VENTURES

Kinetic Intelligence Labs — TITAN FOC v8.2

Founder & Lead Engineer

Oct 2024 – Present

- Founded a robotics hardware venture developing custom field-oriented control (FOC) motor controllers and RL-based training systems for robotic actuators. Targeting the gap between expensive industrial drives and hobbyist-grade boards.
- Designed the TITAN FOC v8.2: a 50.25mm circular 6-layer PCB built around the STM32G474CEU6 MCU and DRV8353RS three-phase gate driver, with six TPW1R306PL MOSFETs, AS5047P magnetic encoder, and TCAN1051 CAN transceiver. 88 components, \$32.39 BOM — fabricated by PCBWay with ENIG finish and 2oz outer copper.
- Architected a 16-sheet hierarchical KiCad schematic following professional template standards with full design validation and interactive BOM generation.
- Exploring university IP commercialization pathways and building go-to-market strategy for the robotic joint actuator and light EV drivetrain segments.

PPO vs. AWR Motion Imitation Research

Cognitive Robotics Lab · 2025 – Present

- Benchmarking Proximal Policy Optimization against Advantage-Weighted Regression for humanoid whole-body motion imitation using NVIDIA IsaacGym physics simulation with MimicKit motion-capture replay.
- Built the full simulation pipeline: WSL2 with CUDA 12 GPU passthrough, conda-managed environments, and custom reward shaping for joint-level tracking accuracy. Research targets comparing sample efficiency, policy robustness, and sim-to-real transfer potential.

LoRa Autonomous RC Boat Fleet

Personal Project · 2024

- Designed and built custom STM32F446-based control boards for a fleet of autonomous surface vehicles with LoRa mesh networking for beyond-line-of-sight communication, u-blox GPS waypoint navigation, CAN bus sensor fusion, and a centralized ground station for real-time telemetry and multi-agent formation control.

Humanoid Robot Simulation

University of Arkansas · 2024

- Built a 24-degree-of-freedom humanoid simulation in ROS2 Humble and Gazebo with a full URDF model, TF transform tree, C++ and Python controller nodes, ros2_control joint trajectory interface, and scripted mission replay for gait cycle analysis.

Robotic Arm Simulator & GUI

University of Arkansas · 2024

- Developed an adjustable-DOF robotic arm simulator with a Tkinter GUI supporting full range-of-motion remote control, real-time inverse and forward kinematics computation, end-effector position tracking, collision detection, and 3D visualization.